

Amirhossein Sourì

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Education

Sharif University of Technology

B.Sc. in Computer Science and Engineering

2022 – Expected June 2027

Tehran, Iran

- GPA: **18.99/20**
- Ranked **14th among 145,000+ participants** in Iran's National University Entrance Examination.

Shahid Beheshti High School (NODET)

Diploma in Mathematics and Physics

2019 – 2022

Tehran, Iran

Affiliated with the National Organization for Development of Exceptional Talents

Research Interests

- Machine and deep learning
- Computer vision and multimodal learning
- Language and vision-language models
- ML Systems
- Generative models
- Trustworthy ML
- Reinforcement learning and multi-agent systems
- Robotics

Research Experience

RIML Lab — Visual Reasoning in Vision-Language Models

July 2026 – Present

Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology

- Developed the project's initial question-conditioned visual-grounding study using CLIP patch/text features, heatmap-to-box conversion, and downstream evaluation with Qwen2.5-VL-3B-Instruct.
- Investigating training and evaluation methods for evidence-sensitive reasoning in vision-language models.

RIML Lab — Diffusion Model Unlearning

Dec 2025 – Mar 2026

Research Assistant (PI: Dr. Mohammad Hossein Rohban), Sharif University of Technology

- Investigated noise-robust concept unlearning for text-to-image diffusion models under adversarial and distributional variation.
- Contributed to adaptive noise sampling and experimental evaluation; this work resulted in a preprint currently under review at ICLR 2027.

Preprint

Weeding Out Bad Seeds: Initial-Noise-Robust Unlearning for Text-to-Image Diffusion Models

[arXiv](#)

Arian Komaei Koma, Seyed Amir Kasaei, Aida Aryafar, Matin Ghiasi, Ali Aghayari, **Amirhossein Sourì**, Mohammad Mosayyebi, AmirMahdi Sadeghzadeh, and Mohammad Hossein Rohban. *Under review at ICLR 2027.*

Teaching Experience

Sharif University of Technology

Oct 2023 – Present

Teaching Assistant

Tehran, Iran

- **Artificial Intelligence** — Dr. Mahdieh Soleymani Fall 2025
- **Artificial Intelligence** — Mr. Samiei and Mr. Fereydooni Spring 2025
- **Machine Learning** — Dr. Abolfazl Motahari Spring 2025
- **Probability and Statistics** — Dr. Amir Najafi Spring 2024
- **Advanced Programming (Java)** — Dr. MohammadAmin Fazli Spring 2024, Spring 2026
- **Fundamentals of Programming (C)** — Dr. MohammadAmin Fazli Fall 2023, Fall 2024, Fall 2025

Industry Experience

Hamravesh

Oct 2024 – Sept 2025

Software Engineer

Tehran, Iran

- Developed and deployed a marketplace service for one-click cloud applications, spanning backend APIs, frontend interfaces, and Kubernetes integration.

Selected Projects

- Question-Conditioned Visual Grounding | [GitHub](#)

July 2026

 - Trained an attention-based model on CLIP patch and question embeddings to localize answer-relevant image regions.
 - Converted predicted heatmaps into bounding boxes and evaluated their effect on downstream VQA with Qwen2.5-VL-3B-Instruct.
- Deep Learning Practical Assignments | [GitHub](#)

Spring 2026

 - Completed 15 implementation-focused notebooks spanning neural networks and optimizers from scratch, CNNs/RNNs, LoRA/QLoRA adaptation, RAG, and language model inference.
 - Implemented and evaluated VAE, DDPM, VQ-VAE-2, SimCLR, CLIP analysis, and latent-diffusion image-editing pipelines using PyTorch.
- Modern Information Retrieval Project | [GitHub](#)

Spring 2026

 - Built a Goodreads search engine with preprocessing, near-duplicate detection, inverted/tiered indexes, BM25 and language model ranking, and MAP/NDCG/MRR evaluation.
 - Developed a hybrid multimodal product-search pipeline combining dense multimodal embeddings, SPLADE++ sparse retrieval, cross-encoder reranking, and structured LLM responses.
- Machine Unlearning and Robustness Exercises | [GitHub](#)

Dec 2025

 - Implemented class unlearning in a conditional VAE using Fisher information and evaluated recovery attacks based on label manipulation, latent interpolation, and condition optimization.
- Artificial Intelligence Practical Assignments | [GitHub](#)

Spring 2024

 - Implemented classical search, constraint satisfaction, minimax with alpha-beta pruning, Bayesian-network inference, HMMs, Q-learning, SARSA, and value iteration.
- Machine Learning Models Collection | [GitHub](#)

Sept 2023

 - Implemented, explained, and compared core supervised and unsupervised methods—including regression, SVMs, trees, ensembles, dimensionality reduction, clustering, and Gaussian mixtures—with scikit-learn baselines.

Technical Skills

Programming	Python, C/C++, Java, TypeScript/JavaScript, SQL, R
Machine Learning	PyTorch, Transformers, Diffusers, PEFT, TRL, Accelerate, scikit-learn, SentenceTransformers, TensorFlow
Data & Vision	NumPy, pandas, Hugging Face Datasets, FAISS, OpenCV, Matplotlib, Jupyter
Systems & Software	Linux, Git, Docker, Kubernetes, vLLM, Django/DRF, React, PostgreSQL
Documentation	LaTeX, Markdown

Selected Coursework

Deep Learning	Dr. Mahdieh Soleymani	19.7/20
Machine Learning	Dr. Ali Sharifi Zarchi	20.0/20
Artificial Intelligence	Dr. Mohammad Hossein Rohban	19.3/20
Modern Information Retrieval	Dr. Mahdieh Soleymani	20.0/20
Computer Simulation	Dr. Bardia Safaei	In progress
Linear Algebra	Dr. Maryam Ramezani	20.0/20
Probability and Statistics	Dr. Amir Najafi	19.2/20

Languages

Persian: Native proficiency **English:** Professional working proficiency

Selected Academic Service & Leadership

Public Relations Officer, CE Students' Scientific Chapter	July 2025 – Present
Senior Editor, Byte Publication	May 2025 – Present
Vice President & Technical Lead, Hardwar Event	Feb 2025 – May 2025